coursera

Menu

Optional programming assignment

In this assignment we are interested in the performance of KD-trees on n uniformly distributed points in the unit square. Below you will find three datasets for $n = 10, n = 100$ and $n = 1000$. Here we will use the query rectangle with lower left corner $(0.5, 0.5)$ and upper right corner $(0.8, 0.8)$, however feel free to experiment with different queries.

UnitSquare10
TXT File

UnitSquare100
TXT File

UnitSquare1000
TXT File

An interesting thing to look at is the number of visited nodes in the KD-tree that did not contribute to the output, i.e. a point in the query range. Finally compare the running time of the KD-tree against a brute force algorithm that loops over all the points to verify whether it is in the query range or not. You can discuss your findings below.

Participation is optional

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